

Resonant phase-root boundary, pure phase-orbit Wick control, and Besov one-use reduction

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claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

first issued 2026-07-24 · this version issued 2026-07-24 · v1.0

1 Purpose and scope

R-073 reduced the unresolved nonlinear part of A13 to adapted terminal phase-root coercivity. It left two possible proof mechanisms: a signed strict-past cancellation or a positive regularity gain for the value-high Taylor branches. This note decides three parts of that fork.

First, the unsubtracted nonlinear secant remainder has a nonzero high-high-to-low production resonance in its mismatched E_x-b_ℓ branch. Its bare low Fourier coefficient is independent of shell separation. Therefore no coefficient-blind theorem can place that bare root in $H^{-1/2+\rho}$ with $\rho > 0$, and R-063's balanced jet gain may not be imported before the resonance and all lower chaoses are identified.

Second, strict-past dependence and phase-root algebra do not automatically center the terminal Wick coefficient. A smooth production-frame diagnostic has a strictly negative expectation. It is not a coercivity counterexample: it rotates a value while freezing its derivative and has no ultraviolet replication defeating Cameron–Martin cost.

Third, a genuine integrable local phase orbit is much better. The full raw current is exactly invariant. The only Wick change is the phase-noncommuting part of the A1 derivative covariance. A resolvent identity makes that part cutoff-uniform with an $O(\Lambda^{-3})$ tail, so the anomaly is paid by one arbitrarily small terminal sextic allocation. Thus the pure phase-orbit subchannel closes; the remaining gate is the adapted gauge-quotient, horizontal Taylor one-form.

The note also proves the deterministic Besov product estimate which would pay that horizontal root once at the raw R-050 order. It needs a sixth moment, not a positive regularity gain. Constructing the corresponding adapted, lower-chaos-complete one-form is still open. This result does not prove terminal coercivity, finite-energy recovery, controlled-shell one-use, the $q = 10/9$ Nelson theorem, an interacting measure, removal of the density floor or regulator, or infinite volume.

All production statements use the hash-pinned A1 coefficients, fixed $\epsilon_\rho > 0$, the six-real Pauli convention, the common real-even regulator class, and the R-066/R-073 finite-low convention.

2 Production frames and notation

Let S_1, S_2, S_3 be the realified Pauli generators, and write

$$q_r(z) = \frac{z^T S_r z}{|z|^2 + e}, \quad p_r(z) = 2S_r z, \quad v_r(z) = 2(S_r - q_r(z)I)z, \quad (2.1)$$

$$M_r(z) = [p_r(z), v_r(z)], \quad B(z) = \sum_{r=1}^3 M_r(z) Q_{\Pi} M_r(z)^T. \quad (2.2)$$

Put

$$P = M_X^2 + \mu_X, \quad Q_{\text{II}} = \begin{pmatrix} q_{11} & q_{12} \\ q_{12} & q_{22} \end{pmatrix}. \quad (2.3)$$

At the production values,

$$q_{12} + q_{22} = \frac{27}{1600P}. \quad (2.4)$$

For a smooth six-real field z , the cutoff covariance-normal current is

$$\mathcal{W}_\Lambda(z) = \frac{1}{2} \sum_i \int B(z) : (D_i z \otimes D_i z - \Gamma_{\Lambda,i}) dx. \quad (2.5)$$

The elementary frame bounds are

$$\|p_r(z)\| \leq 2|z|, \quad \|v_r(z)\| \leq 2|z|, \quad \text{Tr } B(z) \leq 24\lambda_Q |z|^2. \quad (2.6)$$

For the second inequality, expand $|(S_r - q_r I)z|^2$ and use $q_r = (z^T S_r z)/(|z|^2 + e)$; the two q_r terms have a nonpositive net correction to $|S_r z|^2 \leq |z|^2$.

3 Frozen principal resonance

Take $z = e_1$ and $a = G = e_2$. Only the real σ_x frame survives after projection on e_2 . Direct differentiation gives

$$M_x(e_1)^T e_2 = (2, 2), \quad \frac{1}{2} D^2 p_x(e_1)[e_2, e_2] = 0, \quad (3.1)$$

$$\frac{1}{2} D^2 v_x(e_1)[e_2, e_2] = -\frac{4}{1+e} e_2. \quad (3.2)$$

The other two Pauli generators have zero e_2 projection. Equations (2.4), (3.1), and (3.2) therefore yield the exact principal coefficient

$$\boxed{e_2 \cdot \frac{1}{2} \sum_r D^2 M_r(e_1)[e_2, e_2] Q_{\text{II}} M_r(e_1)^T e_2 = -\frac{27}{200P(1+e)}}. \quad (3.3)$$

It equals $-0.0337499999999578\dots$ at the pinned parameters and is strictly nonzero.

There is an exact finite-secant version. On the normalized one-dimensional torus put

$$a_j(x) = t e_2 \cos(kx), \quad G(x) = g e_2 \cos(nx), \quad b_\ell(x) = b e_2 \cos(nx), \quad k > n. \quad (3.4)$$

For $E_x = M_x(e_1 + a_j) - M_x(e_1) - D M_x(e_1)[a_j]$,

$$E_x^p = 0, \quad e_2 \cdot E_x^v = -\frac{4t^2 \cos^2(kx)}{1+e+t^2 \cos^2(kx)}. \quad (3.5)$$

Thus the raw mismatched branch is

$$\Lambda_{k,n} = -8(q_{12} + q_{22})gb \int \frac{t^2 \cos^2(kx)}{1+e+t^2 \cos^2(kx)} \cos^2(nx) d\bar{x}. \quad (3.6)$$

The first factor has Fourier support only at multiples of $2k$. Since $k > n$, its nonzero modes are orthogonal to $\cos(2nx)$, and

$$\boxed{\Lambda_{k,n} = -4(q_{12} + q_{22})gb \left(1 - \sqrt{\frac{1+e}{1+e+t^2}}\right)}. \quad (3.7)$$

It is independent of k/n . The zero Fourier mode of the nonlinear multiplier is nonzero, so multiplying a Gaussian derivative at frequency n retains a positive multiple of its variance. No bare factor $2^{-\rho(j-\ell)}$ and no raw $H^{-1/2+\rho}$ improvement can follow from the production coefficient alone.

This is the exact content of NG-2026-07-24-A13-RAW-BARE-POSITIVE-GAIN-ROOT. It does not rule out Cameron–Martin frequency weights, explicit resonance subtraction, a lower-chaos-complete adapted reconstruction, or the signed R-073 telescope.

4 Automatic adapted centering is false

Let J_D be the doublet phase generator, $z = e_1$, and $y = J_D z = e_4$. For $R_\theta = \exp(\theta J_D)$, exact frame evaluation gives

$$y^T B(R_\theta z) y = \lambda_e \sin^2 \theta, \quad \lambda_e = \frac{3(113e^2 + 136e + 48)}{2000P(1+e)^2} > 0. \quad (4.1)$$

Let $\xi \sim N(0, 1)$ be an earlier derivative innovation and choose the smooth bounded strict-past angle

$$\theta_t(\xi) = \arcsin(te^{-\xi^2/2}), \quad 0 < t < 1. \quad (4.2)$$

If one rotates the terminal value but freezes the derivative carrier, the one-coordinate Wick diagnostic is

$$W = \frac{\lambda_e t^2}{2} e^{-\xi^2} (\xi^2 - 1). \quad (4.3)$$

Since

$$\mathbb{E} e^{-\xi^2} = \frac{1}{\sqrt{3}}, \quad \mathbb{E} \xi^2 e^{-\xi^2} = \frac{1}{3\sqrt{3}}, \quad (4.4)$$

one obtains

$$\boxed{\mathbb{E} W = -\frac{\lambda_e t^2}{3\sqrt{3}} < 0.} \quad (4.5)$$

At the pinned parameters $\lambda_e = 0.0180000000000105 \dots$

Equation (4.5) refutes automatic conditional centering of an adapted terminal coefficient and is registered as NG-2026-07-24-A13-AUTOMATIC-ADAPTED-WICK-CENTERING. It is not a production coercivity counterexample. The frozen derivative in (4.3) is not the derivative of the rotated spatial field, and the example supplies no cutoff sequence whose negative budgeted expectation diverges.

5 Pure phase-orbit Wick theorem

Let J_D, J_S be the commuting doublet and singlet phase generators and

$$R(x) = \exp\{\theta_D(x)J_D + \theta_S(x)J_S\}. \quad (5.1)$$

Both generators commute with every S_r . Hence

$$q_r(Rz) = q_r(z), \quad M_r(Rz) = RM_r(z), \quad (5.2)$$

and the phase roots obey

$$M_r(z)^T J_\alpha z = 0, \quad \text{quad } \alpha \in \{D, S\}. \quad (5.3)$$

Differentiating the actual rotated field gives

$$D_i(Rz) = R \left(D_i z + \sum_\alpha (\partial_i \theta_\alpha) J_\alpha z \right). \quad (5.4)$$

Combining (5.2)–(5.4) proves the pathwise identity

$$\boxed{M_r(Rz)^T D_i(Rz) = M_r(z)^T D_i z.} \quad (5.5)$$

Thus the complete raw terminal current square is exactly invariant under any regular local phase orbit. Randomness of the angles is irrelevant to this pointwise identity; no conditioning, projector, pseudoinverse,

or Malliavin derivative is used. An additive displacement $Rz - z$ need not itself be an admissible strict-past control, so (5.5) closes a geometric subchannel rather than the whole control class.

For the Wick term, average a covariance over the compact phase group:

$$\Gamma_{\Lambda,i}^{\parallel} = \int_{\mathbb{T}^2} R_{\vartheta}^T \Gamma_{\Lambda,i} R_{\vartheta} d\vartheta, \quad \Gamma_{\Lambda,i}^{\perp} = \Gamma_{\Lambda,i} - \Gamma_{\Lambda,i}^{\parallel}. \quad (5.6)$$

The parallel part commutes with every phase. Using $B(Rz) = RB(z)R^T$ and (5.5),

$$\boxed{\mathcal{W}_{\Lambda}(Rz) - \mathcal{W}_{\Lambda}(z) = -\frac{1}{2} \sum_i \int \text{Tr}\{[RB(z)R^T - B(z)]\Gamma_{\Lambda,i}^{\perp}\} dx.} \quad (5.7)$$

The A1 Fourier symbol is

$$A(k) = K(k)I + H, \quad K(k) = r + Z|k|^2 + Y|k|^4. \quad (5.8)$$

The scalar resolvent is phase commuting, and

$$(KI + H)^{-1} - K^{-1}I = -K^{-1}(KI + H)^{-1}H. \quad (5.9)$$

For sufficiently large $|k|$, (5.9) gives

$$\|P_{\perp}A(k)^{-1}\| \lesssim |k|^{-8}. \quad (5.10)$$

The derivative covariance contributes a k_i^2 weight. Therefore

$$k_i^2 \|P_{\perp}A(k)^{-1}\| \lesssim |k|^{-6}, \quad \sum_{|k|>\Lambda} |k|^{-6} \lesssim \Lambda^{-3}. \quad (5.11)$$

Consequently

$$\boxed{\sup_{\Lambda} \|\Gamma_{\Lambda,i}^{\perp}\| < \infty, \quad \|\Gamma_{\infty,i}^{\perp} - \Gamma_{\Lambda,i}^{\perp}\| \lesssim \Lambda^{-3}.} \quad (5.12)$$

All covariance norms in (5.12)–(5.13) are operator norms. Equations (2.6), (5.7), and (5.12) yield

$$|\mathcal{W}_{\Lambda}(Rz) - \mathcal{W}_{\Lambda}(z)| \leq 72\lambda_Q \max_i \|\Gamma_{\Lambda,i}^{\perp}\| \|z\|_2^2 \leq \zeta \|z\|_6^6 + C_{\zeta}. \quad (5.13)$$

The terminal sextic is spent once. The common phase $J_D + J_S$ commutes with the entire complex A1 covariance, so its anomaly is exactly zero. Only relative doublet–singlet phase sees the finite defect. In the R-070 telescope the finite-low boundary $-\mathcal{W}_{j_0}$ remains intact; (5.13) does not reset or discard it.

6 Cameron–Martin rescue boundary

The R-071 terminal-kernel fixture has the exact affine slope

$$c_e = \frac{27(6e^2 + 22e + 27)}{400(e + 3)^3} > 0. \quad (6.1)$$

Its periodic production lift has averaged leakage $(c_e/2)t + O(N^{-2})$ and oscillatory H^2 cost

$$\frac{3t^2}{2}(N^2 + 1 + N^{-2}). \quad (6.2)$$

After adding an η Cameron–Martin allocation and optimizing in t ,

$$\inf_t \left\{ \frac{c_e}{2}t + \frac{3\eta}{2}(N^2 + 1 + N^{-2})t^2 \right\} = -\frac{c_e^2}{24\eta(N^2 + 1 + N^{-2})}. \quad (6.3)$$

This is $O_\eta(N^{-2})$ and is dyadically summable.

The same distinction applies to (3.7). Realizing $t \cos(kx)$ through the A1 Cameron–Martin map costs at least the fourth-order frequency weight $ct^2 \langle k \rangle^4$; realizing the low derivative $b \cos(nx)$ costs a further positive low-shell energy. Thus the lack of a *bare coefficient* separation gain is not a failure of the Cameron–Martin-weighted coercivity theorem. Any fixed finite-mode negative diagnostic is also absorbed into C_{η, ζ, j_0} . A genuine falsifier would require a cutoff sequence with unbounded negative budgeted expectation. No registered witness has that property.

7 Besov one-use discriminator

The resonance should not be estimated by the R-073 termwise shell norms. After the exact endpoint grouping, its deterministic payload is cubic in a total control and has the form $A \otimes A \otimes DA$. The correct duality is the endpoint Besov pairing, not a fictitious positive regularity gain.

Lemma 1 (cubic control payload). *For $0 < \kappa < 1/2$ and $A \in H^2(\mathbb{T}^3) \cap L^6(\mathbb{T}^3)$,*

$$\boxed{\|A \otimes A \otimes DA\|_{B_{1,1}^{1/2+\kappa}} \leq C_\kappa \|A\|_{H^2} \|A\|_6^2.} \quad (7.1)$$

Proof. Use an inhomogeneous Littlewood–Paley partition and expand the three factors by largest input frequency. If the differentiated factor has largest scale 2^j , put it in L^2 and the other two factors in L^4 ; the weighted sum is bounded by

$$\|A\|_6^2 \sum_j 2^{(3/2+\kappa)j} \|\Delta_j A\|_2.$$

If an undifferentiated factor has the largest scale, put that factor in L^2 , the other value factor in L^6 , and use $\|DS_j A\|_3 \lesssim 2^j \|A\|_6$ for the lower factor. Comparable-largest terms are assigned to either largest factor and obey the same bound by the vector-valued square-function inequality. Finally Cauchy–Schwarz against $\sum_j 2^{4j} \|\Delta_j A\|_2^2$ converges because $3/2 + \kappa < 2$. The finite low block is controlled by the inhomogeneous norm. Summing the finitely many Bony orientations proves (7.1). $\square$

Let $\mathfrak{J}$ be a reconstructed adapted Taylor one-form at the honest R-050 order

$$\mathfrak{J} \in C^{-1/2-\kappa} = B_{\infty,\infty}^{-1/2-\kappa}. \quad (7.2)$$

Besov duality and (7.1), with $X = \|A\|_{H^2}^2$ and $Y = \|A\|_6^6$, give

$$|\langle \mathfrak{J}, A^2 DA \rangle| \leq C \|\mathfrak{J}\| X^{1/2} Y^{1/3}. \quad (7.3)$$

The exponents 2, 3, 6 are conjugate. Weighted Young therefore yields

$$\boxed{|\langle \mathfrak{J}, A^2 DA \rangle| \leq \eta X + \zeta Y + C \eta^{-3} \zeta^{-2} \|\mathfrak{J}\|^6.} \quad (7.4)$$

There is positive probability slack 1/6. R-050 supplies every finite moment for its unshifted current, so the required sixth moment is compatible with the existing enhancement. What remains unproved is that the complete R-073 horizontal Taylor root, including the resonance and all R-063 lower chaoses, reconstructs as such an adapted $\mathfrak{J}$ with a cutoff-uniform sixth moment and with the terminal square and finite-low boundary retained.

8 Finite-energy recovery is not automatic

Once a uniform regular one-shot control inequality is proved, stopping and smooth cylindrical approximation are natural candidates for extending it. Plain lower semicontinuity or Fatou, however, has the wrong direction for a lower bound if approximants carry excess positive sextic or raw-current energy. Predictability also forbids a terminal-event truncation.

The nontrivial finite-energy successor must construct predictable recovery sequences converging simultaneously in the Cameron–Martin norm, terminal L^6 , and the positive raw-current graph functional (or prove a biting and uniform-integrability substitute). At each fixed cutoff the negative part has the useful bound

$$\mathcal{W}_J(z)^- \leq C_J(1 + \|z\|_2^2), \quad (8.1)$$

but (8.1) alone does not control excess positive energy. Accordingly this note records the density shortcut as unresolved rather than claiming the finite-energy extension.

9 Evidence map and successor

The proof map after R-074 is:

1. **Closed negatively:** the raw unsubtracted positive-gain root is impossible by (3.3)–(3.7). The obstruction is an exact production high–high-to-low resonance, not a fitted slope.
2. **Closed negatively:** strict-past dependence plus Wick ordering does not imply exact centering; (4.1)–(4.5) give a smooth production diagnostic.
3. **Closed positively:** a genuine local phase orbit preserves the complete raw current exactly. Its relative-phase Wick anomaly is cutoff-uniform and potential-grade by (5.7)–(5.13); the common phase anomaly vanishes exactly.
4. **Closed as a deterministic payment:** an order $C^{-1/2-\kappa}$ adapted current pairs with the cubic control payload using one H^2 payment, one sextic payment, and a sixth random moment by (7.1)–(7.4). No positive gain is needed for this pairing.
5. **Not a counterexample:** the terminal-kernel and frozen-resonance witnesses acquire summable or stronger Cameron–Martin costs, as in (6.3).
6. **Still open:** construct the lower-chaos-complete adapted gauge-quotient Taylor one-form, prove its signed endpoint reconstruction and sixth-moment bound, then establish a predictable graph-norm recovery theorem.

The smallest honest successor is

$$\boxed{\text{A13-CLASSII-ADAPTED-GAUGE-QUOTIENT-TAYLOR-ONE-FORM.}} \quad (9.1)$$

It must keep the R-073 terminal square, both restored first variations, the explicit resonant low output, every R-063 lower chaos, and the finite-low boundary. It may close either by the direct Besov payment (7.4) or by a resonance-subtracted positive gain. It may not claim gain for the complete raw root.

10 Devil’s-advocate review

1. **“R-063 already gives the needed adapted gain.” UPHELD as false.** Equation (3.7) retains a nonzero unbalanced low mode at arbitrary separation. R-063 concerns balanced unshifted jets and explicitly excludes the adapted two-control theorem.

2. **“No bare gain means coercivity is false.” DISMISSED.** The actual controls carry fourth-order Cameron–Martin weights. Equations (6.2)–(6.3) show the strongest known kernel lift is saved by a summable loss.
3. **“Strict-past adaptation centers the terminal Wick form.” UPHELD as false.** Equation (4.5) is strictly negative. Stein or Clark–Ocone repair would introduce undeclared Malliavin derivatives of the control.
4. **“The negative phase diagnostic contradicts gauge invariance.” DISMISSED.** The diagnostic rotates the value and freezes its derivative. A true spatial phase differentiates according to (5.4), and its raw current is exactly invariant by (5.5).
5. **“The Wick covariance spoils the pure phase theorem.” DISMISSED with a finite mitigation.** The commuting covariance part cancels. The remaining relative-phase block has the summable resolvent tail (5.11), and (5.13) pays it once from the sextic.
6. **“The Besov lemma proves terminal coercivity.” UPHELD as an overclaim.** The deterministic payment is closed, but the complete adapted horizontal one-form and its sixth moment have not been constructed.
7. **“Ordinary L^2 density proves the finite-energy extension.” UPHELD as false.** Lower semicontinuity acts in the wrong direction for excess positive raw energy. A predictable graph-norm recovery theorem is required.
8. **“The phase projector may replace the signed terminal completion.” UPHELD as false.** R-073’s rank jumps and restored first variations remain binding. The present phase-orbit theorem uses no projector and does not alter the R-073 identity.
9. **“This result proves one-use or Nelson.” UPHELD as an overclaim.** Horizontal adapted coercivity, finite-energy recovery, controlled-shell one-use, and Nelson remain open.

11 Reproduction and independent audit

The primary executable derives Q_{II} , e , and P from A1; evaluates all three six-real frames; checks the exact principal coefficient, finite secant integral, value-only phase diagnostic, local phase covariance, phase-group covariance tail, Cameron–Martin scaling, Besov exponent ledger, and the quadratic coefficient bound. The non-importing executable rebuilds the complex Pauli representation, sharp-cube covariance, finite-difference principal, quadrature, phase orbit, and tail independently. The integrated verifier hash-pins R-050, R-063, R-073, the sources, note, PDF, and manifest before executing both children.

```
python codes/foundations/
  a13_classii_resonant_phase_root_besov_reduction_verify.py
```

Result footer

Result ID: A13-CLASSII-RESONANT-PHASE-ROOT-BESOV-REDUCTION

Precise statement: The unsubtracted nonlinear secant remainder has an exact nondecaying high-high-to-low bare-coefficient resonance in its mismatched branch; automatic adapted Wick centering is false; genuine local phase orbits are raw-invariant with a UV-finite potential-grade Wick anomaly; and an order $-1/2$ -current pays the cubic control payload by a sixth moment Besov one-use inequality.

Scope: Fixed-floor A1/R-050/R-063/R-073; common real-even cutoffs; regular local phase orbits; one-shot strict-past successor class; finite-low retained.

Dependencies: A1; R-050; R-063; R-066; R-069--R-073.

Evidence grade: ANALYTIC, EXACT, EXECUTED.

Reproduction command: python codes/foundations/a13_classii_resonant_phase_root_besov_reduction_verify.py

Expected output: primary + independent + integrated aggregate PASS.

Falsification gate: A principal, secant, phase, covariance-tail, Besov-exponent, hash, PDF, or scope assertion fails.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: No horizontal adapted coercivity, finite-energy recovery, controlled-shell one-use, Nelson, measure, removal, infinite volume, or T5--T7.

Next required action: Construct the adapted gauge-quotient Taylor one-form and close it by the direct Besov payment or an explicitly resonance-subtracted gain.